

Alyssa Mazak

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Education & Technical Skills

Messiah University

Mechanicsburg, PA

BS Biomedical Engineering; Mechanical Engineering Minor

2022 – 2026

- Enrolled in Honors Program
- GPA 3.89
- **Relevant Coursework:** Mechanical Design, Design of Medical Devices, Materials Engineering, Circuits, Fluid Dynamics, Thermodynamics, Robotics, Biomedical Laboratory Techniques
- **Technical:** MATLAB, Arduino, SolidWorks, MS Excel, Word, Power Point

Certifications

- Passed Fundamentals of Engineering Exam, Mechanical (Completed: April 2026)
- OSHA 10-hour Construction Safety and Health Training (Completed: June 2025)
- Certified SOLIDWORKS Associate (Completed: April 2023)
- UR e-Series Application, Pro, and Core Track Training (Completed: January 2026)

Engineering Experience

Voith – Hydro Division

York, PA

Field Service Intern

June 2025 – August 2025

- Designed and analyzed custom lifting and securing systems for turbine refurbishment
- Performed structural, stress, and stability analyses on custom made devices
- Led design reviews and incorporated feedback into revised models and calculations
- Documented device designs, analyses, and implementation methods for client and team review
- Compared bolt vs. weld fasteners through stress analysis and recommended optimal connection methods
- Skills learned: Solid Edge, hand stress calculations, client communication, technical presentation skills

Collaboratory – Messiah University

Mechanicsburg, PA

Stance Control Orthosis - Project Team Member

August 2023 – May 2026

- Team Goal: Design an affordable, fully locking (at any angle) modular knee orthosis
- Programmed algorithms to characterize gait events and compile results to quantify brace response time and reliability in preparation for clinical trials
- Constructed a comprehensive clinical trial plan with detailed experimental design and inclusion criteria
- Redesigned the orthosis' locking mechanism in SolidWorks for improved strain relief and ease of use
- Manufactured and created assembly instructions for the locking mechanism's housing and solenoid
- Conducted project reviews with clients
- Skills learned: MATLAB, professional documentation, technical writing, collaboration

Relevant Experience

Additional Projects

- **F.O.R.C.E Implemented Crutch:** Developed a load-sensing device that measures force applied to assistive walking aids and translates data into a real-time dependency score (August – December 2025)
- **Ball Balancing Robot:** Engineered a feedback-controlled self-balancing robotic platform that uses overhead vision tracking to drive real-time actuator-based stability corrections (January – May 2026)

Kuflik Dermatology

Toms River, NJ

Medical Assistant

May 2024 – September 2024

- Prepared patients for procedures by obtaining informed consent and completing all pre-procedure setups
- Assisted physicians in examinations and procedures, ensuring efficient workflow and patient comfort
- Maintained clean, organized, and fully stocked exam rooms to support continuous clinical operations
- Conducted thorough patient interviews, documenting medical histories and current health concerns
- Skills learned: conflict resolution, patient care, multitasking in high stress environments